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Theory(Logic)

This is a C++ program that shows how to use multiple inheritance. We create 3 three classes: Car, Boat, and DualModeVehicle. Car class has a function drive() that prints "driving on land". Boat class has a function sail() that prints "sailing on water". DualModeVehicle class inherits from both Car and Boat. This is called multiple inheritance. DualModeVehicle class has a function use() that calls drive() from the Car class and sail() from the Boat class. In the main() function: An object myVehicle of type DualModeVehicle is created. The use() function is called using this object. As a result, it prints both messages one for driving, one for sailing.

Pseudo Code

- **Step 1:** Include input-output library
- **Step 2:** Use standard namespace
- **Step 3:** Create class Car
 - **Public:**
 - Function drive()
 - print "driving on land"
- **Step 4:** Create class Boat
 - **Public:**
 - Function sail()
 - print "sailing on water"
- **Step 5:** Create class DualModeVehicle that inherits from Car and Boat
 - **Public:**
 - Function use()
 - call drive()
 - call sail()
- **Step 6:** Start main function
 - Create object myVehicle of DualModeVehicle
 - Call use() using myVehicle
- **Step 7:** End main function

Program Code

```
#include<iostream>
using namespace std;
class Car
{
    public:
    void drive()
    {
        cout<<"driving on land"<<endl;
    }
};

class Boat
{
    public:
    void sail()
    {
        cout<<"sailing on water"<<endl;
    }
};

class DualModeVehicle: public Car,public Boat
{
    public:
    void use()
    {
        drive();
        sail();
    }
};
```

```
int main()
{
    DualModeVehicle myVehicle;
    myVehicle.use();
    return 0;
}
```

Output

```
driving on land  
sailing on water
```

Conclusion

Code is run successfully, required output is received.